

A Complete Sector Classification for Geometrically Regular Weighted Shifts

Automated mathematics run `fa_banach.001`, lane 10

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Status. Candidate full solution, likely valid, pending human review. All three parts of Conjecture 1.3 of Benhida–Curto–Exner are proved and strengthened to four global if-and-only-if classifications.

1 Source conjecture

For $p > 1$ and $-1 < N, D < 1$, Benhida, Curto, and Exner associate the geometrically regular weighted shift W_α with squared weights

$$a_n := \alpha_n^2 = \frac{p^n + N}{p^n + D}, \quad n \geq 0. \tag{1}$$

Their Conjecture 1.3 asks whether: every nonboundary point of Sector II fails Bernstein interpolation; every nonboundary point of Sector III fails moment infinite divisibility; and the listed loci exhaust all $\mathcal{MID}$, subnormal, and completely hyperexpansive shifts in the parameter square [1].

The source says “no α is interpolated” in the first clause. Its immediately preceding theorem and the discussion following Theorem 2.5 make clear that this refers to the squared-weight sequence $a = (\alpha_n^2)$; we follow that established convention.

that is,

$$\alpha_n := \sqrt{\frac{p^n + N}{p^n + D}}, \quad n = 0, 1, 2, \dots$$

Then W_α has the following properties in the sectors given in the diagram:

- (1) Along the main diagonal $D = N$, W_α is the (unweighted) unilateral shift and thereby both $\mathcal{MID}$ and completely hyperexpansive;
- (2) In Sector I, the weight squared sequence α is interpolated by a Bernstein function and W_α is $\mathcal{MID}$;
- (3) In Sector II, the weight squared sequence α is interpolated by a log Bernstein function and W_α is $\mathcal{MID}$, and at least one shift in this sector does not have α interpolated by a Bernstein function;
- (4) In Sector III, W_α is subnormal, and at least one shift in this sector is not $\mathcal{MID}$;
- (5) In Sector IV, W_α is subnormal with finitely atomic Berger measure along the rays $D = p^k N$, $k = 0, 1, 2, \dots$, and is not $\mathcal{MID}$ except for $N = D$;
- (6) In Sector VIIIA, W_α is completely hyperexpansive.

We note as well the following conjectures as suggested by the above.

Conjecture 1.3. Let $p > 1$ and (N, D) such that $-1 < N < 1$ and $-1 < D < 1$. Let W_α be a weighted shift with weight sequence α , where

$$\alpha_n := \sqrt{\frac{p^n + N}{p^n + D}}, \quad n = 0, 1, 2, \dots$$

Then W_α has the following properties in the sectors given in the diagram:

- (1) In Sector II, except on the boundary with Sector I, no α is interpolated by a Bernstein function;
- (2) In Sector III, W_α is subnormal, and except on the boundary with Sector II, no shift is $\mathcal{MID}$;
- (3) Other than as specified above, there are no $\mathcal{MID}$, subnormal, or completely hyperexpansive shifts in the square.

Figure 1: Full-width source crop from arXiv:2309.05888v2, PDF page 7, showing the proved sector list and all of Conjecture 1.3.

2 Proof intuition

Put $q = p^{-1} \in (0, 1)$. The two difficult sequence conditions become signed atomic Hausdorff moment expansions supported at $q, q^2, \dots$. In Sector II, the complement $1 - a_n$ has a negative mass at q^2 , preventing complete alternation and hence Bernstein interpolation. In Sector III, $-\log a_n$ has negative masses at every even atom, preventing the log complete alternation equivalent to $\mathcal{MID}$. Compact Hausdorff moment measures are unique, so these negative masses cannot be hidden by another positive representation.

For complete hyperexpansivity, the first-difference moment sequence must be completely monotone and therefore log-convex. Its exact successive ratio is known from the source. Requiring those ratios to increase forces $D \geq pN$, which cuts out precisely the sector the source already proved sufficient. The source's Sector-IV determinant theorem and the elementary monotonicity of the original weights then finish the global subnormal and $\mathcal{MID}$ classifications.

3 Statement of the complete classification

Theorem 1. *Fix $p > 1$ and $-1 < N, D < 1$, and define W_α by (1). Then:*

1. *The squared-weight sequence (a_n) is interpolated by a Bernstein function if and only if*

$$D = N \quad \text{or} \quad N < D \leq 0.$$

2. *W_α is moment infinitely divisible if and only if*

$$D = N \quad \text{or} \quad N < 0 \quad \text{and} \quad N < D \leq -N.$$

3. *W_α is subnormal if and only if one of the following holds:*

$$\begin{aligned} D = N; \quad N \leq 0 \quad \text{and} \quad D > N; \\ 0 < N < D \quad \text{and} \quad D = p^k N \quad \text{for some } k \in \mathbb{N}. \end{aligned}$$

4. *W_α is completely hyperexpansive if and only if*

$$D = N \quad \text{or} \quad N < 0 \quad \text{and} \quad pN \leq D < N.$$

Consequently all three clauses of Conjecture 1.3 in [1] hold.

4 A signed Hausdorff uniqueness lemma

We use the forward difference convention $\nabla x_n = x_n - x_{n+1}$. A sequence is completely monotone if $\nabla^j x_n \geq 0$ for all $j, n \geq 0$; it is completely alternating if $\nabla^j x_n \leq 0$ for all $j \geq 1$ and $n \geq 0$.

Lemma 1 (negative-atom obstruction). *Let $r_k \in [0, 1]$ be distinct and let $\sum_k |c_k| < \infty$. If*

$$x_n = \sum_k c_k r_k^n \quad (n \geq 0)$$

is completely monotone, then every $c_k \geq 0$.

Proof. By the Hausdorff moment theorem [4, Chapter III], complete monotonicity gives a finite positive measure μ on $[0, 1]$ with moments x_n . The finite signed measure $\nu = \sum_k c_k \delta_{r_k}$ has the same moments. Hence $\mu - \nu$ annihilates every polynomial. Polynomials are uniformly dense in $C([0, 1])$, so the Riesz representation theorem implies $\mu - \nu = 0$. Thus $\nu = \mu$ is positive, and every atomic coefficient is nonnegative. $\square$

This compact signed-measure uniqueness principle is also emphasized in the later Berger-charge work of the same authors [3].

5 Bernstein interpolation

The sign of the first weight difference follows directly from (1):

$$a_{n+1} - a_n = \frac{q^n(1-q)(D-N)}{(1+Dq^n)(1+Dq^{n+1})}. \quad (2)$$

If $D < N$, the sequence decreases, so it cannot be sampled from a Bernstein function, which is increasing. If $D = N$, it is the constant sequence one.

It remains to treat $N < D$. Suppose first that $D > 0$. Since $|D| < 1$,

$$1 - a_n = \frac{(D-N)q^n}{1+Dq^n} \quad (3)$$

$$= \sum_{k=1}^{\infty} (D-N)(-D)^{k-1}(q^k)^n. \quad (4)$$

This is an absolutely summable signed atomic representation. Its coefficient at q^2 is $-(D-N)D < 0$. Were (a_n) completely alternating, the nonnegative sequence $(1 - a_n)$ would be completely monotone, contradicting Lemma 1. Samples of a Bernstein function are completely alternating, so no Bernstein interpolant exists.

For $N < D \leq 0$, the source constructs a Bernstein interpolant [1, Corollary 2.6]. These cases and the constant diagonal prove part (1) of Theorem 1. In particular, throughout Sector II away from its $D = 0$ boundary, $D > 0$ and Bernstein interpolation fails. This proves Conjecture 1.3(1), and (3) also rules out such interpolation in Sectors III and IV.

6 Moment infinite divisibility

For a contractive weighted shift, the source recalls that $\mathcal{MID}$ is equivalent to log complete alternation of the squared weights. When $a_n \leq 1$, this is equivalent to complete monotonicity of

$$t_n = -\log a_n.$$

In Sector III write $N = -u$ and $D = v$, where $0 \leq u < v < 1$. Absolute convergence of the logarithmic series gives

$$t_n = \log \left(\frac{1 + vq^n}{1 - uq^n} \right) \quad (5)$$

$$= \sum_{k=1}^{\infty} \frac{u^k + (-1)^{k+1}v^k}{k} (q^k)^n. \quad (6)$$

For every even k the coefficient is $(u^k - v^k)/k < 0$. Lemma 1 shows that (t_n) is not completely monotone. Hence no interior Sector-III shift is $\mathcal{MID}$.

On the boundary $v = u$, all even coefficients in (5) vanish and the odd coefficients are positive, consistently with the source's Sector-II $\mathcal{MID}$ theorem. Thus Conjecture 1.3(2) holds, including the edge $u = 0 < v$.

To get the global equivalence in Theorem 1(2), combine this obstruction with the source's $\mathcal{MID}$ results in Sectors I and II, its non- $\mathcal{MID}$ result for the subnormal Sector-IV rays [1, Theorem 2.9], and the fact that $\mathcal{MID}$ implies subnormality.

7 Subnormality

A subnormal weighted shift is hyponormal, and a unilateral weighted shift is hyponormal exactly when its weights are nondecreasing. Thus (2) excludes every point with $D < N$, except the constant diagonal.

For the remaining half of the square, the source proves subnormality in Sectors I–III, including a countably atomic Berger measure for Sector III [1, Theorem 2.8]. In Sector IV, Theorem 2.9 proves subnormality on the rays $D = p^k N$, $k \geq 1$. Theorem 2.16 proves that every point off these rays fails some finite k -hyponormality condition and hence cannot be subnormal. Algebraically, these are exactly the alternatives in part (3) of Theorem 1.

8 Complete hyperexpansivity

Let (γ_n) be the moment sequence of W_α , so $\gamma_{n+1} = a_n \gamma_n$, and put

$$\delta_n = \gamma_{n+1} - \gamma_n.$$

Complete hyperexpansivity is equivalent to complete monotonicity of (δ_n) [1, Theorem 2.10 and its preliminaries]. If $D > N$, then $a_0 < 1$ and $\delta_0 < 0$, which is impossible. The diagonal is the unweighted shift, so assume $D < N$; then every $\delta_n > 0$.

Every positive completely monotone sequence is a Hausdorff moment sequence and is log-convex by Cauchy–Schwarz. Consequently the ratios $r_n = \delta_{n+1}/\delta_n$ must be nondecreasing. Directly (and as computed in the source),

$$r_n = \frac{1}{p} \frac{p^n + N}{p^n + D/p} = q \frac{1 + Nq^n}{1 + Dq^{n+1}}. \quad (7)$$

A second exact subtraction gives

$$r_{n+1} - r_n = \frac{q^{n+1}(1 - q)(Dq - N)}{(1 + Dq^{n+1})(1 + Dq^{n+2})}. \quad (8)$$

Therefore log-convexity forces $Dq - N \geq 0$, equivalently $D \geq pN$. Together with $D < N$, this is possible only when $N < 0$ and

$$pN \leq D < N.$$

This is precisely Sector VIIIA. Source Theorem 2.10 proves the entire sector completely hyperexpansive, including its boundary. Adding the diagonal proves part (4) of Theorem 1 and supplies the last missing necessity statement in Conjecture 1.3(3).

9 Assembly, verification, and novelty

The four equivalences cover every point in the open parameter square. The only special boundaries are transparent in the signed expansions: $D = 0$ leaves a positive one-atom measure in (3); $D = -N$ deletes every even atom in (5); and $D = pN < N < 0$ makes (8) an equality. The positive Sector-IV rays and the diagonal are separately covered by the source theorems cited above.

The bundled script `code/verify_q_series.py` symbolically verifies (2), the coefficients in (3)–(5), and (8). It is an algebra check, not a replacement for the Hausdorff uniqueness proof.

A bounded novelty search on 13 August 2026 covered local run indexes, the exact title and conjecture label, and web/arXiv combinations of the sector and property names. The later safe-quotients paper [2] and signed-charge paper [3] repeat only the partial classification; the 2026-updated source v2 still states Conjecture 1.3. No later claimed solution or these exact necessity arguments were found. Novelty confidence is therefore moderate, not definitive.

Human review should focus on Lemma 1, the contractive MID equivalence used before (5), and the log-convexity necessity leading to (8). No unproved dependency is currently known.

References

- [1] C. Benhida, R. E. Curto, and G. R. Exner, *Geometrically regular weighted shifts*, arXiv:2309.05888v2.
- [2] C. Benhida, R. E. Curto, and G. R. Exner, *MID and subnormal safe quotients for geometrically regular weighted shifts*, J. Math. Anal. Appl. **538** (2024), 128443; arXiv:2312.06390.
- [3] C. Benhida, R. E. Curto, and G. R. Exner, *Signed representing measures (Berger-type charges) in subnormality and related properties of weighted shifts*, arXiv:2405.15000; published in Results in Mathematics (2025).
- [4] N. I. Akhiezer, *The Classical Moment Problem and Some Related Questions in Analysis*, Oliver and Boyd, 1965.